UNIT – III

Equilibrium, Free Body Diagram, Lami's Theorem and Beams

Equilibrium

A body is said to be in **equilibrium** when the **resultant of all the forces acting on it is zero**. In this condition, the body **does not accelerate** and remains either **at rest** or **moves with constant velocity**.

In simple words, equilibrium means **all forces acting on the body are balanced**.

Types of Equilibrium

1. Static Equilibrium

When a body is **at rest** and the resultant force is zero.

Example

- A book resting on a table
- A wall clock hanging on a nail
- A bridge carrying stationary load

In these cases, forces are balanced and the body **does not move**.

2. Dynamic Equilibrium

When a body is **moving with constant velocity** but the resultant force is zero.

Example

- A car moving on a straight road with constant speed
- A lift moving upward with constant velocity

Even though the body is moving, forces are balanced, so it is still in equilibrium.

Conditions of Equilibrium

For a **particle in a plane**, two conditions must be satisfied.

First Condition

Sum of all horizontal forces must be zero.

$$\Sigma F_x = 0$$

This means **forces acting to the right = forces acting to the left**.

Second Condition

Sum of all vertical forces must be zero.

$$\Sigma F_y = 0$$

This means **upward forces = downward forces**.

For Rigid Bodies (Additional Condition)

For a rigid body, **moment equilibrium** must also be satisfied.

$$\Sigma M = 0$$

This means **clockwise moment = anticlockwise moment**.

Example of Equilibrium

Book on a Table

Forces acting on the book:

1. **Weight (W)** acting downward due to gravity
2. **Reaction force (R)** from the table acting upward

```
  R
  ↑
 ┌──────┐
 │ Book │
 └──────┘
  ↓
  W
```

Since R = W

The resultant force becomes **zero**, so the book remains **at rest**.

Importance of Equilibrium in Engineering

The concept of equilibrium is widely used in:

- Structural engineering
- Bridge design
- Machine design
- Building construction
- Crane and cable systems

Engineers use equilibrium to ensure **structures remain stable and safe**.

Equilibrant

The **equilibrant** is the force that **balances the resultant force of all other forces acting on a body**.

It is the force required to **bring a body into equilibrium**.

Characteristics of Equilibrant

The equilibrant has three important properties:

1. **Equal magnitude** to the resultant force
2. **Opposite direction** to the resultant
3. **Same line of action (collinear)** as the resultant

Thus, it cancels the resultant force completely.

Relation between Resultant and Equilibrant

If resultant force is (R), then equilibrant force is E = -R. This negative sign indicates **opposite direction**.

Example

Suppose two forces act on a body and produce a **resultant force of 50 N toward the right**.

To bring the body into equilibrium, an equilibrant must act:

- Magnitude = **50 N**
- Direction = **toward the left**

50 N → (Resultant)
50 N ← (Equilibrant)

Both forces cancel each other.

Resultant = **0**

Therefore, the body becomes **balanced**.

Practical Engineering Example

Suspended Load with Rope

Consider a load hanging from a rope.

Forces acting:

1. **Weight (W)** downward
2. **Tension (T)** upward

Tension (T)
↑
|
●
|
↓
Weight (W)

If T = W, then system is in equilibrium.

Here **tension acts as the equilibrant** balancing the weight.

Difference between Resultant and Equilibrant

Resultant	Equilibrant
Single force representing combined effect of all forces	Force that balances the resultant
Acts in direction of net force	Acts opposite to resultant
Causes motion if unbalanced	Prevents motion
Symbol: R	Symbol: E

Free Body and Free Body Diagram (FBD)

Free Body

A **free body** is a body that is **isolated from all surrounding objects** so that the **forces acting on it can be studied clearly**.

When a body is separated from its surroundings, the **supports and connections are removed**, and the forces acting on those supports are replaced by **reaction forces**.

Purpose of Studying a Free Body

- To understand **how forces act on a body**
- To apply **equilibrium conditions**
- To calculate **unknown forces or reactions**

Examples of Free Bodies

- A block on a table
- A beam carrying loads
- A hanging weight attached to a rope
- A ladder resting against a wall

In each case, the body is **separated from the environment** for analysis.

Free Body Diagram (FBD)

A **Free Body Diagram (FBD)** is a **simple sketch that shows a body isolated from its surroundings along with all external forces acting on it**.

It is one of the **most important tools in engineering mechanics** because it helps in solving equilibrium problems.

Forces Shown in an FBD

The following forces are usually shown:

1. **Weight (W)** – acting vertically downward due to gravity
2. **Normal Reaction (R)** – force exerted by the surface
3. **Tension (T)** – force in a rope, cable, or string
4. **Applied Force (P)** – external force applied to the body
5. **Friction Force (F)** – force opposing motion
6. **Support Reactions** – forces from supports like hinges or rollers

Steps to Draw a Free Body Diagram

Step 1: Identify the Body

Select the body or object that needs to be analyzed.

Step 2: Isolate the Body

Separate the body from its surroundings.

Step 3: Remove Supports

All supports or contacts with other bodies are removed.

Step 4: Replace Supports with Reaction Forces

Replace each support with the **appropriate reaction force**.

Examples:

- Roller → vertical reaction
- Hinge → horizontal and vertical reactions
- Rope → tension force

Step 5: Show All External Forces

Draw all forces acting on the body with proper direction.

Example: Block on a Horizontal Surface

Forces acting on the block:

1. **Weight (W)** acting downward

2. **Reaction (R)** from the surface acting upward

```
  R
  ↑
┌──────────┐
│ Block    │
└──────────┘
  ↓
  W
```

Where:

- **W = Weight of the block**
- **R = Reaction force from the surface**

If R = W, Then the body remains in **equilibrium**.

Importance of Free Body Diagram

Free body diagrams are used to:

- Solve **equilibrium problems**
- Determine **support reactions**
- Analyze **forces in structures**
- Study **mechanical systems**

Almost every problem in **engineering mechanics begins with an FBD**.

3. Methods of Analyzing Equilibrium

When a body is in equilibrium, the forces acting on it must satisfy certain conditions.

Two main methods are used to analyze equilibrium:

1. **Analytical Method**
2. **Graphical Method**

1. Analytical Method

The **analytical method** uses **mathematical equations and calculations** to determine unknown forces.

This method is widely used in **engineering design and analysis** because it gives **accurate numerical results**.

Conditions of Equilibrium

For a body in a plane:

$$\Sigma F_x = 0$$

$$\Sigma F_y = 0$$

For rigid bodies:

$$\Sigma M = 0$$

Where:

- (∑Fx) = Sum of horizontal forces
- (∑Fy) = Sum of vertical forces
- (∑ M) = Sum of moments about a point

Procedure of Analytical Method

1. Draw the **Free Body Diagram**
2. Resolve inclined forces into **horizontal and vertical components**
3. Apply equilibrium equations
4. Solve the equations to find unknown forces

Example

A body is acted upon by three forces:

- 10 N toward right
- 6 N toward left
- 4 N upward

Resolve forces:

Horizontal direction:

10 − 6 = 4 N

Vertical direction:

4 N upward

To maintain equilibrium, equal forces must act in **opposite directions**.

Advantages of Analytical Method

- Highly **accurate**
- Suitable for **complex engineering problems**
- Can be applied to **large force systems**
- Provides **exact numerical solutions**

2. Graphical Method

The **graphical method** determines the magnitude and direction of forces using **scale drawings and vector diagrams**.

Instead of equations, **force diagrams** are constructed.

Principle

For a body in equilibrium, the **force vectors form a closed polygon**.

If the polygon closes, the **resultant force is zero**, and the body is in equilibrium.

Common Graphical Methods

1. Parallelogram Law of Forces

Used when **two forces act at a point**.

The resultant is represented by the **diagonal of the parallelogram**.

2. Triangle Law of Forces

Used when **three forces act at a point**.

If the three forces can be represented by the **three sides of a triangle taken in order**, the body is in equilibrium.

3. Polygon Law of Forces

Used when **more than three forces act on a body**.

For equilibrium, the **force polygon must form a closed shape**.

Example

If three forces acting on a body form a **closed triangle**, the body is in equilibrium.

```
    F1
   /  \
  /    \
 F2------F3
```

Since the triangle closes, the resultant force is **zero**.

Lami's Theorem

Lami's Theorem is an important principle used to solve problems involving **three forces acting at a point in equilibrium**.

It provides a **simple mathematical relationship between the magnitudes of the forces and the angles between them**.

The theorem is widely used in **engineering mechanics** to determine **unknown forces in cables, ropes, and structural members**.

If a **body is in equilibrium under the action of three concurrent forces**, then each force is **proportional to the sine of the angle between the other two forces**.

In other words, the **ratio of each force to the sine of the angle between the other two forces is constant**.

Mathematical Expression

$$\frac{F_1}{\sin\alpha} = \frac{F_2}{\sin\beta} = \frac{F_3}{\sin\gamma}$$

Where:

- F_1, F_2, F_3 = three forces acting at a point
- α = angle between F_2 and F_3
- β = angle between F_1 and F_3
- γ = angle between F_1 and F_2

Thus, each force is proportional to the sine of the angle between the other two forces.

Explanation of Lami's Theorem

Consider a particle **O** acted upon by three forces:

- (T1)
- (T2)
- (W)

All forces meet at the **same point**.

```
   T1
   /
  /
 O
/ \
T2  W
```

Where:

- **T1 and T2** are tensions in strings or cables
- **W** is the weight acting vertically downward

If the system is in equilibrium:

- The **three forces form a closed triangle of forces**.
- Therefore, their magnitudes follow **Lami's relation**.

This allows us to calculate **unknown tensions or forces easily**.

Conditions for Applying Lami's Theorem

Lami's theorem is valid only when the following conditions are satisfied:

1. Body Must Be in Equilibrium
The resultant of all forces must be **zero**.

2. Only Three Forces Must Act
Exactly **three forces** should act on the body.

3. Forces Must Be Concurrent
All forces must **intersect at a single point**.

4. Forces Must Be Coplanar
All forces must lie in the **same plane**.

If these conditions are not satisfied, **Lami's theorem cannot be applied**.

Engineering Applications of Lami's Theorem
Lami's theorem is commonly used in many engineering situations where **three forces act at a point**.

1. Suspended Loads

Used to determine **tension in supporting cables**.

Example:
- A load hanging by two ropes.

2. Cable Systems

Used in **guy wires, suspension bridges, and cable supports**.

3. Crane Hooks

When a crane lifts a load using **two cables**, Lami's theorem helps calculate the **tension in each cable**.

4. Ring with Three Forces

If three strings pull a **ring**, their tensions can be calculated using Lami's theorem.

5. Bridge Joints

Used in **truss structures** to determine **forces in members meeting at a joint**.

Advantages of Lami's Theorem

- Simplifies **three-force equilibrium problems**
- Reduces the need for **complex calculations**
- Useful for **quick engineering analysis**

Limitations of Lami's Theorem

- Applicable **only for three forces**
- Forces must be **concurrent**
- Not suitable for **multi-force systems**

Beam

A **beam** is a **structural member usually placed horizontally that is designed to carry loads acting perpendicular to its axis**.

The beam **supports loads and transfers them safely to the supports or columns**.

Beams mainly resist **bending and shear forces** caused by loads.

Functions of a Beam

The main functions of a beam are:

- To **support loads**
- To **transfer loads to supports**
- To **maintain structural stability**
- To **distribute loads over a structure**

Examples of Beams in Engineering

Beams are widely used in many structures such as:

- **Bridges**
- **Building floors and roofs**
- **Machine frames**
- **Railway tracks**
- **Crane structures**

These beams help in **carrying loads safely without failure**.

Loads Acting on Beams

The loads acting on a beam may be:

- **Point load**
- **Uniformly distributed load (UDL)**
- **Inclined load**
- **Couple or moment**

These loads produce **bending and shear stresses** in the beam.

Types of Beams

Beams are classified based on the **type of support provided at their ends**.

The main types of beams are:

1. Simply Supported Beam
2. Cantilever Beam
3. Overhanging Beam
4. Fixed Beam

1. Simply Supported Beam

A **simply supported beam** is a beam that is **supported at both ends**, usually by a **hinge support and a roller support**.

One end allows **rotation**, and the other allows **horizontal movement**.

Characteristics

- One **hinge (pin) support**
- One **roller support**
- Beam can **rotate slightly at supports**
- No moment resistance at supports
- Most commonly used beam in structures

Applications

Simply supported beams are used in:

- **Bridges**
- **Building floors**
- **Roof supports**
- **Structural frames**

2. Cantilever Beam

A **cantilever beam** is a beam that is **fixed at one end and free at the other end**.

The fixed end resists **vertical force, horizontal force, and moment**.

Fixed end Free end

Characteristics

- One end **rigidly fixed**
- Other end **free**
- Fixed end resists **bending moment**
- Large **bending stress occurs near the fixed end**

Applications

Cantilever beams are commonly used in:

- **Balconies**
- **Crane arms**
- **Street light poles**
- **Aircraft wings**
- **Cantilever bridges**

3. Overhanging Beam

An **overhanging beam** is a beam that **extends beyond one or both supports**.

Here the beam **extends beyond the support**, forming an **overhang**.

Characteristics

- Supported at two points
- Beam extends **beyond one or both supports**
- Load may act on the **overhanging part**

Applications

Overhanging beams are used in:

- **Balconies**
- **Roof projections**
- **Canopies**
- **Bridge structures**

4. Fixed Beam
A **fixed beam** is a beam whose **both ends are rigidly fixed into the supports**.
The supports **prevent rotation and movement** of the beam.

Fixed Fixed

Characteristics

- Both ends are **fixed**
- Rotation at supports is **not allowed**
- Supports resist **moment and forces**
- Beam has **higher strength and rigidity**

Applications

Fixed beams are used in:

- **Rigid frame structures**
- **Reinforced concrete buildings**
- **Machine structures**
- **Structural frames**

Types of Supports

In structural engineering, **supports** are devices or structural components that **hold a beam in position and transfer loads from the beam to the foundation or ground**.

Supports **restrict certain movements of the beam** and provide **reaction forces** that keep the structure in equilibrium.

Different supports allow or prevent **translation and rotation** of the beam.

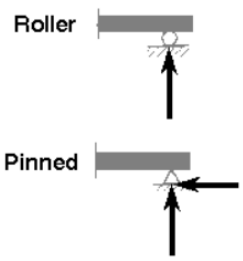
The three most common types of supports are:

1. Hinged (Pin) Support
2. Roller Support
3. Fixed Support

1. Hinged (Pin) Support

A **hinged support**, also called a **pin support**, is a support that **prevents horizontal and vertical movement but allows rotation** of the beam.

It provides **two reaction forces** but **no moment reaction**.



Reactions Provided

A hinged support provides:

- **Horizontal reaction (Hx)**
- **Vertical reaction (Vy)**

But it **cannot resist moment**, so the beam is free to rotate at the support.

Characteristics

- Allows **rotation of the beam**
- Prevents **horizontal movement**
- Prevents **vertical movement**
- Provides **two reaction components**

Practical Examples

Hinged supports are used in:

- **Bridge structures**
- **Roof trusses**
- **Structural frames**
- **Door hinges**

2. Roller Support

A **roller support** is a support that **prevents vertical movement but allows horizontal movement and rotation**.

It provides **only one reaction force**, which is usually **vertical**.

Reactions Provided

A roller support provides:

- **One vertical reaction force**

It **does not resist horizontal force or moment**.

Characteristics

- Allows **horizontal movement**
- Allows **rotation**
- Prevents **vertical movement**
- Provides **one reaction only**

Purpose of Roller Supports

Roller supports are used to **allow expansion and contraction of beams due to temperature changes**.

If both ends were fixed, thermal expansion could cause **large stresses and damage**.

Applications

Roller supports are commonly used in:

- **Bridges**
- **Long beams**
- **Steel structures**
- **Railway bridges**

3. Fixed Support

A **fixed support** is a support that **completely restrains the beam**.

It prevents:

- Horizontal movement
- Vertical movement
- Rotation

Therefore, it provides **three reactions**.

Reactions Provided

A fixed support provides:

1. **Horizontal reaction**
2. **Vertical reaction**
3. **Moment reaction**

Thus, the beam **cannot move or rotate at the fixed end**.

Characteristics

- Prevents **all movements**
- Provides **maximum stability**
- Produces **bending moment at support**

Applications

Fixed supports are used in:

- **Cantilever beams**
- **Building columns**
- **Rigid frame structures**
- **Machine foundations**

Types of Loads Acting on Beam

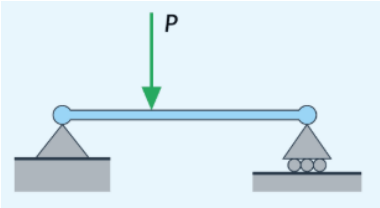
In structural engineering, **loads** are forces that act on a beam and cause **bending, shear, or rotation**.
These loads are transferred from the beam to the **supports and foundation**.

Loads acting on beams may come from:

- Weight of the structure itself
- People or vehicles
- Machines and equipment
- Wind or environmental forces

Understanding the types of loads helps engineers **design safe and stable structures**.

1. Point Load (Concentrated Load)
A **point load** is a load that acts at a **single point on the beam**.
Although in reality loads are applied over a small area, in engineering calculations they are **assumed to act at one point**.



Characteristics

- Acts at **one specific point**
- Represented by a **single arrow**
- Unit is **kN or N**

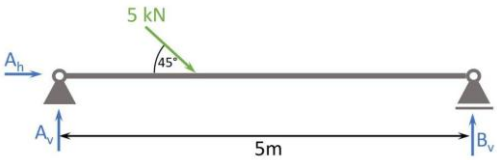
Examples

- Weight of a **machine placed on a beam**
- **Wheel load of a vehicle on a bridge**
- **Column load transferred to a beam**

2. Inclined Load

An **inclined load** is force acting on a beam at an **angle other than 90°** to the beam.

Such a force must be **resolved into horizontal and vertical components** to analyze the beam.



Components of Inclined Load

An inclined force **P** can be resolved into:

- **Horizontal component (Px)**
- **Vertical component (Py)**

Examples

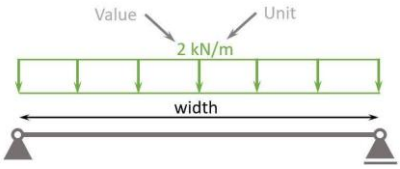
Inclined loads occur in:

- **Cable-supported beams**
- **Truss structures**
- **Wind acting on structures**

3. Uniformly Distributed Load (UDL)

A **Uniformly Distributed Load (UDL)** is a load that is **spread evenly over a certain length of the beam**.

The load intensity is **constant throughout the length**.



Example

If a load of **2 kN/m acts over a beam of length 4 m**:

Total load: = 2 * 4 = 8 kN
This load acts at the **center of the loaded length**.

Examples in Engineering

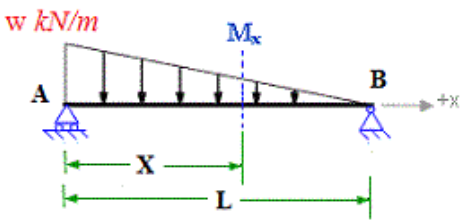
UDL occurs due to:

- **Weight of a slab on a beam**
- **Uniform floor loads**
- **Distributed material loads**

4. Uniformly Varying Load (UVL)

A **Uniformly Varying Load (UVL)** is a load whose **intensity changes uniformly along the length of the beam**.

The load distribution usually forms a **triangular shape**.



Resultant of Triangular Load
The **total load** acting on the beam is equal to the **area of the triangle**.
Total load:

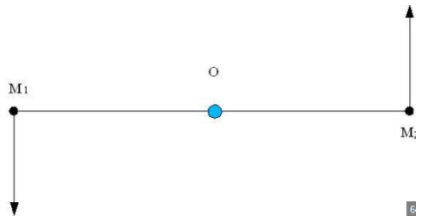
$$W = \frac{1}{2} wL$$

Where:
- w = maximum load intensity (kN/m)
- L = length of beam (m)

The resultant force acts at **one-third of the base from the larger intensity side**.

5. Couple (Moment)

A **couple** consists of **two equal and opposite parallel forces acting at a distance apart**.
A couple **does not produce translation** but causes **rotation of the beam**.



Moment or couple is measured in:
kN·m (kilonewton meter)

Characteristics
- Produces **rotation only**
- Does **not cause linear motion**
- Depends on **force magnitude and distance between forces**

Moment of a couple:

M = F*d

Where:
- **M** = Moment
- **F** = Force
- **d** = distance between forces

Examples
Couples occur in:
- **Wrenches and spanners**
- **Steering wheels**
- **Rotating shafts**
- **Mechanical torque systems**